

MINNIPA, Australia

WMO: 956620

Lat: 32.8428S

Lon: 135.1514E

Elev: 545

StdP: 14.41

Time Zone: 9.50 (AUA)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.9	41.5	16.9	13.2	78.0	20.8	15.9	79.7	27.3	57.0	24.9	58.1	7.6	130	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	103.7	65.0	99.7	63.9	95.6	63.0	70.0	86.0	68.1	86.7	66.4	85.3	13.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	97.8	73.2	63.3	88.9	71.5	60.9	81.4	69.9	34.3	84.2	32.7	86.4	31.3	84.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	22.3	20.2	DB	36.2	110.8	1.7	2.9	34.9	112.9	34.0	114.6	33.0	116.2	31.8	118.4	(4)
(5)			WB	34.9	72.8	1.6	2.6	33.8	74.7	32.9	76.2	32.0	77.7	30.9	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	75.2	74.4	70.1	65.2	58.5	53.4	52.4	53.8	58.1	63.4	68.0	71.2	(6)	
	DBStd	10.46	8.43	8.16	7.67	6.56	4.98	3.55	3.64	4.78	6.66	7.67	8.42	8.56	(7)	
	HDD50	35	0	0	0	0	0	7	14	11	3	0	0	0	(8)	
	HDD65	1862	5	7	29	76	212	349	392	349	229	127	60	27	(9)	
	CDD50	4993	781	684	624	455	265	109	87	129	247	415	539	658	(10)	
	CDD65	1341	320	271	187	82	12	1	0	1	22	77	149	219	(11)	
	CDH74	17283	4321	3358	2137	791	59	2	1	15	298	1128	2091	3082	(12)	
	CDH80	9078	2532	1873	1044	276	5	0	0	1	81	483	1073	1710	(13)	
(14) Wind	WSAvg	11.4	13.1	12.6	11.8	10.7	9.9	9.7	10.4	10.8	11.2	11.8	12.1	12.5	(14)	
(15) Precipitation	PrecAvg	12.3	0.5	0.6	0.7	0.7	1.1	1.7	1.7	1.6	1.3	0.9	0.7	0.9	(15)	
	PrecMax	20.9	3.1	2.4	2.5	1.9	2.9	3.6	3.5	3.2	2.9	3.3	1.9	3.9	(16)	
	PrecMin	5.6	0.0	0.0	0.0	0.0	0.1	0.5	0.5	0.2	0.1	0.0	0.0	0.0	(17)	
	PrecStd	2.9	0.6	0.6	0.7	0.5	0.6	0.8	0.7	0.9	0.8	0.9	0.5	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.2	106.5	101.5	93.6	79.8	70.5	70.1	76.1	88.0	95.8	102.6	106.7	(19)	
		MCWB	66.8	67.7	64.1	61.9	57.4	54.5	53.1	54.4	58.1	60.4	62.5	64.4	(20)	
	2%	DB	104.4	101.5	95.5	87.1	74.9	65.7	65.2	71.2	81.7	90.5	96.6	100.8	(21)	
		MCWB	65.6	65.7	62.9	59.2	55.6	52.3	51.6	53.2	56.1	58.7	61.8	63.3	(22)	
	5%	DB	99.1	97.0	90.6	82.3	71.0	62.7	62.0	67.2	77.0	85.3	91.3	95.7	(23)	
		MCWB	64.4	64.7	62.0	58.1	54.7	52.1	51.6	52.2	55.0	57.2	60.2	61.9	(24)	
	10%	DB	93.7	91.5	85.7	78.2	67.3	60.1	59.3	63.2	72.0	80.3	86.0	90.2	(25)	
		MCWB	63.6	63.7	61.2	57.0	54.0	51.9	50.8	50.9	53.8	55.8	59.2	61.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.1	72.9	70.3	65.7	61.1	59.0	57.1	58.0	61.2	63.7	68.2	70.3	(27)	
		MCDB	87.6	89.6	83.3	74.4	69.9	65.1	60.9	68.5	77.3	81.6	81.9	83.2	(28)	
	2%	WB	69.3	70.4	67.4	62.7	59.2	56.2	54.7	55.9	58.8	61.2	64.8	67.3	(29)	
		MCDB	89.6	86.3	81.6	76.2	66.7	60.3	60.5	63.7	73.5	79.4	85.7	84.5	(30)	
	5%	WB	67.5	68.2	65.4	61.0	57.8	54.7	53.3	54.4	56.9	59.4	62.9	65.0	(31)	
		MCDB	90.1	85.8	80.2	74.1	64.9	58.6	58.8	62.0	71.2	77.5	84.4	85.3	(32)	
	10%	WB	65.7	66.0	63.6	59.5	56.4	53.3	52.1	52.8	55.3	57.8	61.1	63.1	(33)	
		MCDB	87.4	84.6	79.1	71.6	63.7	57.7	57.4	60.5	68.1	74.7	80.8	84.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	27.9	26.3	23.7	21.0	16.5	14.3	14.7	17.9	22.3	25.9	27.0	27.2	(35)	
		MCDBR	34.5	32.0	29.5	26.5	20.6	17.1	18.0	23.0	29.8	33.3	33.3	34.4	(36)	
	5% WB	MCWBR	9.3	8.9	8.4	8.5	8.4	8.8	9.3	10.5	10.8	10.9	10.2	10.1	(37)	
		MCWBR	27.5	25.1	23.6	20.8	16.1	13.2	15.3	18.7	25.7	27.9	28.6	27.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.339	0.339	0.320	0.312	0.300	0.292	0.293	0.300	0.327	0.337	0.341	0.330	(40)	
		taud	2.518	2.536	2.571	2.579	2.595	2.603	2.593	2.560	2.473	2.447	2.468	2.525	(41)	
	Ebn at Noon		317	311	305	291	279	274	279	291	298	307	314	321	(42)	
		Edh at Noon	36	34	30	27	23	22	23	27	33	36	37	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2500	2245	1847	1368	990	840	905	1197	1622	2030	2313	2480	(44)	
	RadStd		122	120	106	85	45	39	40	70	107	134	107	120	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+108	

Nomenclature: See separate page